

Notes: Peng and Xiong (JFE, 2006)

Investor Attention, Overconfidence and Category Learning

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1	Big picture
•	Question. How does limited investor attention shape what investors learn about firms, and what does that imply for return comovement and return predictability?
•	Setting. The paper builds a dynamic equilibrium model with a representative investor, many risky assets, and three layers of fundamentals: market, sector, and firm-specific. The investor has a scarce attention budget that must be allocated across these different sources of uncertainty.
•	Core challenge. Investors do not passively observe fundamentals. They must process information, and cognitive capacity is limited. Because attention to one task necessarily crowds out attention to another, the information structure itself is endogenous.
•	Main conceptual contribution. The paper shows that limited attention generates category learning : investors optimally process relatively more market and sector information than firm-specific information. Stocks within a sector are therefore learned about partly as a category rather than purely as separate firms.
•	Second conceptual contribution. When category learning is combined with overconfidence —modeled as overestimating the precision of self-processed information—the model generates excess return comovement and cross-sectional patterns of predictability that standard rational-expectations models struggle to explain.
•	Main equilibrium result. Asset prices are linear in the investor’s conditional expectation of next-period dividends. The investor chooses attention allocations to minimize the conditional variance of portfolio dividends next period.
•	Main category-learning result. If common factors are quantitatively important enough, the

investor allocates more attention to the market factor than to a sector factor, and more to a sector factor than to a firm-specific factor. Under severe attention constraints, firm-specific information is ignored altogether.

- **Main return-variance result.** With unbiased Bayesian updating, more information resolves uncertainty earlier and does not raise return variance. With enough overconfidence and information below a threshold, more processed information can **raise** return variance because investors overreact to noisy signals.
- **Main comovement result.** Return correlations can exceed the underlying correlations of fundamentals because investors overweight common information relative to firm-specific information.
- **Main cross-sectional result.** Sectors with more efficient information processing have lower average return correlation and more informative prices, because investors treat firms in those sectors less as one category and more as separate firms.
- **Main time-series result.** As information technology improves over time, investors allocate relatively more attention to firm-specific information, so firm-specific volatility rises and average return correlation falls.
- **Main predictability result.** After controlling for overconfidence, sectors with lower average return correlation should display stronger overreaction-driven firm-level predictability, but weaker predictive power of ignored public signals.
- **Economic takeaway.** The paper's big idea is that comovement is not only about common fundamentals. It is also about *how investors choose to learn*. Limited attention shifts learning toward categories, and overconfidence amplifies the pricing effects of that shift.

2 Model environment and key objects

2.1 Basic environment

- Time is discrete and infinite: $t = 1, 2, \dots$
- There is one representative investor, many risky assets, and one risk-free asset paying constant gross return $1 + r$.
- The investor has exponential utility:

$$u(c) = -\frac{1}{\gamma}e^{-\gamma c}, \quad (1)$$

and chooses consumption to maximize

$$\max E_t \left[\sum_{s=t}^{\infty} \delta^{s-t} u(c_s) \right]. \quad (2)$$

Interpretation. Exponential utility and Gaussian uncertainty deliver a linear equilibrium and keep the attention-allocation problem tractable.

2.2 Factor structure of fundamentals

The investor holds a portfolio spanning m sectors, each with n firms. The dividend of firm j in sector i at time t is

$$d_{i,j,t} = h_t + f_{i,t} + g_{i,j,t}, \quad (3)$$

where:

- h_t is the **market factor**,
- $f_{i,t}$ is the **sector factor**,
- $g_{i,j,t}$ is the **firm-specific factor**.

These are independent across dimensions and across time, with Gaussian distributions:

$$h_t \sim N(\bar{h}, \sigma_h^2), \quad f_{i,t} \sim N(\bar{f}, \sigma_f^2), \quad g_{i,j,t} \sim N(\bar{g}, \sigma_g^2). \quad (4)$$

Interpretation. The model isolates exactly where common movement can come from: either from common fundamentals themselves or from the way investors allocate attention across these layers of fundamentals.

2.3 Attention allocations

The investor allocates a total attention budget across the market factor, each sector factor, and each firm-specific factor. Let

$$L_t = (\lambda_{h,t}, \lambda_{f,1,t}, \dots, \lambda_{f,m,t}, \lambda_{g,1,1,t}, \dots, \lambda_{g,m,n,t})', \quad (5)$$

with the constraint

$$\lambda_{h,t} + \sum_{i=1}^m \lambda_{f,i,t} + \sum_{i=1}^m \sum_{j=1}^n \lambda_{g,i,j,t} \leq 1. \quad (6)$$

Interpretation. Attention is scarce. Allocating more to one factor necessarily means allocating less to others.

2.4 Signals and information processing

The investor processes information and obtains noisy signals about next period's fundamentals:

$$s_{h,t} = h_{t+1} + \varepsilon_{h,t}, \quad (7)$$

$$s_{f,i,t} = f_{i,t+1} + \varepsilon_{f,i,t}, \quad (8)$$

$$s_{g,i,j,t} = g_{i,j,t+1} + \varepsilon_{g,i,j,t}. \quad (9)$$

Signal noise variances depend on attention. Using entropy, the paper measures the amount of information in a processed signal as a linear function of attention. For the market factor,

$$I(h_{t+1}; s_{h,t}) = \frac{1}{2} y_h k \lambda_{h,t}, \quad (10)$$

where k is total attention capacity and y_h is the efficiency of processing market information. For sector and firm-specific factors, the corresponding perceived information amounts are

$$\hat{I}(f_{i,t+1}; s_{f,i,t}) = \frac{1}{2} \phi y_i k \lambda_{f,i,t}, \quad \hat{I}(g_{i,j,t+1}; s_{g,i,j,t}) = \frac{1}{2} \phi y_i k \lambda_{g,i,j,t}. \quad (11)$$

Here $\phi \geq 1$ is the **overconfidence parameter**. If $\phi = 1$, the investor is unbiased. If $\phi > 1$, the investor exaggerates the precision of self-processed information.

Interpretation. k says how much total cognitive capacity the investor has. y_i says how productive that capacity is in a given sector. ϕ says how strongly the investor overreacts to the information she thinks she has extracted.

2.5 Posterior beliefs

Given the processed signals, the investor updates beliefs by Bayes' rule. For the market factor:

$$\hat{h}_t = \bar{h} + \left(1 - e^{-\lambda_{h,t} y_h \phi k}\right) (s_{h,t} - \bar{h}), \quad (12)$$

$$\omega_{h,t}^2 = \sigma_h^2 e^{-\lambda_{h,t} y_h \phi k}. \quad (13)$$

Similarly, for sector and firm-specific factors:

$$\hat{f}_{i,t} = \bar{f} + \left(1 - e^{-\lambda_{f,i,t} y_i \phi k}\right) (s_{f,i,t} - \bar{f}), \quad \omega_{f,i,t}^2 = \sigma_f^2 e^{-\lambda_{f,i,t} y_i \phi k}, \quad (14)$$

$$\hat{g}_{i,j,t} = \bar{g} + \left(1 - e^{-\lambda_{g,i,j,t} y_i \phi k}\right) (s_{g,i,j,t} - \bar{g}), \quad \omega_{g,i,j,t}^2 = \sigma_g^2 e^{-\lambda_{g,i,j,t} y_i \phi k}. \quad (15)$$

Interpretation. More attention lowers posterior variance. Overconfidence makes the response coefficient $1 - e^{-\lambda y \phi k}$ too large, so prices overreact to processed signals.

2.6 Information efficiency and cross-sector heterogeneity

The paper allows sectors to differ in y_i , the efficiency with which investors transform attention into usable information.

- Higher y_i can reflect better disclosure, clearer fundamentals, richer analyst coverage, or lower complexity.
- The paper lets the same sector-level efficiency parameter apply both to sector factors and to firms inside that sector.

Interpretation. Cross-sector differences in y_i are the model's bridge from theory to cross-sectional empirical predictions about return synchronicity and price informativeness.

2.7 Key return objects

The stock price of firm (i, j) is $p_{i,j,t}$. Excess return is

$$q_{i,j,t} = p_{i,j,t} + d_{i,j,t} - (1 + r)p_{i,j,t-1}. \quad (16)$$

Two correlation objects matter later:

$$\rho = \text{Corr}(q_{i,j,t}, q_{k,l,t}), \quad i \neq k, \quad (17)$$

and

$$\rho_i = \text{Corr}(q_{i,j,t}, q_{i,k,t}), \quad j \neq k. \quad (18)$$

Interpretation. ρ is cross-sector comovement; ρ_i is within-sector comovement. The theory explains both.

3 Models and analytical specifications (all equations + interpretation)

3.1 Dynamic optimization and equilibrium pricing

The investor's Bellman equation is

$$J_t(K_t) = \max_{L_t} E_t \left[\max_{c_t} (u(c_t) + \delta E_t[J_{t+1}(K_{t+1}) \mid S_t]) \right], \quad (19)$$

with next-period cash holdings

$$K_{t+1} = (K_t - c_t)(1 + r) + \sum_{i=1}^m \sum_{j=1}^n d_{i,j,t+1}. \quad (20)$$

Asset prices satisfy the standard Euler equation,

$$P_t = E_t \left[\delta \frac{u'(c_{t+1})}{u'(c_t)} (P_{t+1} + D_{t+1}) \right]. \quad (21)$$

The paper's linear equilibrium price function is

$$P_t = \frac{1}{1+r} (E(D_{t+1} \mid S_t) + L_t), \quad (22)$$

where L_t is a deterministic risk-premium vector.

Interpretation. Prices move with the investor's processed information because conditional expectations move with posterior beliefs.

3.2 Theorem 1: the investor's attention problem

The paper shows that the investor chooses attention to solve

$$V_t = \min_{L_t} \text{Var}_t \left(\sum_{i=1}^m \sum_{j=1}^n d_{i,j,t+1} \mid S_t \right). \quad (23)$$

Using the factor structure and the posterior variances, this becomes

$$V_t = \min_{L_t} \left[m^2 n^2 \sigma_h^2 e^{-\lambda_{h,t} y_h \phi k} + \sum_{i=1}^m n^2 \sigma_f^2 e^{-\lambda_{f,i,t} y_i \phi k} + \sum_{i=1}^m \sum_{j=1}^n \sigma_g^2 e^{-\lambda_{g,i,j,t} y_i \phi k} \right]. \quad (24)$$

Interpretation. Attention is allocated where it reduces *portfolio-level* uncertainty the most. A market factor matters for all firms, a sector factor matters for all firms in one sector, and a firm-specific factor matters for only one firm. This is the direct mathematical source of category learning.

3.3 Theorem 2: existence and first-order conditions

The attention-allocation problem has a unique solution. The first-order conditions equate the marginal benefit of attention to a common opportunity-cost multiplier $\mu_{k,t}$:

$$\mu_{k,t} = y_h \phi k (m^2 n^2 \sigma_h^2) e^{-\lambda_{h,t} y_h \phi k} \quad \text{if } \lambda_{h,t} > 0, \quad (25)$$

$$\mu_{k,t} = y_i \phi k (n^2 \sigma_f^2) e^{-\lambda_{f,i,t} y_i \phi k} \quad \text{if } \lambda_{f,i,t} > 0, \quad (26)$$

$$\mu_{k,t} = y_i \phi k \sigma_g^2 e^{-\lambda_{g,i,j,t} y_i \phi k} \quad \text{if } \lambda_{g,i,j,t} > 0. \quad (27)$$

Interpretation. The multiplier $\mu_{k,t}$ is the shadow value of one unit of attention. A factor receives positive attention if and only if its marginal uncertainty reduction justifies this shadow value.

3.4 Proposition 1: category learning

Under symmetric sectors and firms, if

$$m^2 \sigma_h^2 > \sigma_f^2 \quad \text{and} \quad n^2 \sigma_f^2 > \sigma_g^2, \quad (28)$$

then the investor allocates more attention to the market factor than to a sector factor, and more to a sector factor than to a firm-specific factor.

In the severely constrained case,

$$k < \frac{m}{\phi y} \log \left(\frac{n^2 \sigma_f^2}{\sigma_g^2} \right), \quad (29)$$

the investor allocates attention only to market and sector factors and ignores firm-specific data.

Interpretation. The investor rationally prefers information that helps with many securities at once. Limited attention therefore generates endogenous category behavior even when categories are not imposed exogenously.

3.5 Return variance decomposition and Proposition 2

Return variance decomposes into market, sector, and firm-specific pieces:

$$\text{Var}(q_{i,j,t}) = \text{Var} \left(\frac{1}{1+r} \hat{h}_t + h_t - \hat{h}_{t-1} \right) + \text{Var} \left(\frac{1}{1+r} \hat{f}_{i,t} + f_{i,t} - \hat{f}_{i,t-1} \right) + \text{Var} \left(\frac{1}{1+r} \hat{g}_{i,j,t} + g_{i,j,t} - \hat{g}_{i,j,t-1} \right). \quad (30)$$

For a generic factor, the paper writes the variance contribution as

$$\text{Var} \left(\frac{1}{1+r} \hat{f}_{i,t} + f_{i,t} - \hat{f}_{i,t-1} \right) = u(x, \phi) \sigma_f^2, \quad (31)$$

where $x = \lambda y k$ is the amount of information processed and $u(x, \phi)$ is the return-variance amplification function.

Proposition 2 has two parts:

- If $\phi = 1$, more information resolves uncertainty earlier and therefore does not raise return variance; with positive discounting it lowers return variance.
- If the investor is sufficiently overconfident,

$$\phi > 1 + \frac{(1+r)^2 - 1}{(1+r)^2 + 1}, \quad (32)$$

and the amount of processed information is below a threshold, then more information and more overconfidence *increase* return variance.

Interpretation. This is the paper's key non-rational force. More information is not always stabilizing because an overconfident investor overreacts to the signal she extracts.

3.6 Proposition 3: excess comovement

For firms in different sectors, only the market component is shared:

$$\rho = \frac{\text{Var} \left(\frac{1}{1+r} \hat{h}_t + h_t - \hat{h}_{t-1} \right)}{\sqrt{\text{Var}(q_{i,j,t}) \text{Var}(q_{k,l,t})}}. \quad (33)$$

For firms in the same sector, both market and sector components are shared:

$$\rho_i = \frac{\text{Var} \left(\frac{1}{1+r} \hat{h}_t + h_t - \hat{h}_{t-1} \right) + \text{Var} \left(\frac{1}{1+r} \hat{f}_{i,t} + f_{i,t} - \hat{f}_{i,t-1} \right)}{\sqrt{\text{Var}(q_{i,j,t}) \text{Var}(q_{i,k,t})}}. \quad (34)$$

Under the conditions of Proposition 2B and Proposition 1, return correlations exceed the corresponding correlations of fundamentals.

Interpretation. Investors overreact more to common information than to firm-specific information, so prices move together more than cash-flow fundamentals do.

3.7 Proposition 4: comovement and price informativeness

The paper measures a firm's price informativeness by the remaining uncertainty in future payoffs:

$$\text{Remaining uncertainty} = \text{Var}(h_t - \hat{h}_{t-1}) + \text{Var}(f_{i,t} - \hat{f}_{i,t-1}) + \text{Var}(g_{i,j,t} - \hat{g}_{i,j,t-1}). \quad (35)$$

Consider two sectors with different information-efficiency parameters, $y_1 > y_2$. Then the investor processes more total information in sector 1 and also relatively more firm-specific information there.

Under the variance-amplification region of Proposition 2B, sector 1 has:

- more informative prices, and
- lower average return correlation.

Interpretation. Better information-processing technology makes investors less reliant on category-level learning and more willing to distinguish firms inside the sector.

3.8 Proposition 5: time trend in correlation

If all factors share a common information-efficiency parameter y and y increases over time, then investors allocate relatively more attention to firm-specific information. Consequently:

- firm-specific return volatility rises relative to sector-level volatility, and
- return correlation between firms falls over time.

Interpretation. Advances in information technology relax effective attention constraints and make stock-price movements more idiosyncratic.

3.9 Proposition 6: overreaction-driven predictability

If $\phi > 1$, the first-order autocovariance of a firm's stock return is negative. Under Proposition 2B, this negative autocovariance in firm-specific returns becomes stronger when the investor processes more firm-specific information.

Interpretation. Overreaction creates reversals. But the reversals show up most strongly where investors actually attend to firm-specific information.

3.10 Proposition 7: predictive power of ignored information

The paper adds an ignored public signal about the firm-specific factor:

$$\nu_{i,j,t} = g_{i,j,t+1} + \varepsilon_{\nu,i,j,t}. \quad (36)$$

Proposition 7 shows that the return predictive power of this ignored signal decreases with the amount of firm-specific information the investor has already processed.

Interpretation. Ignored information is valuable only when investors have left a lot of firm-specific uncertainty unresolved.

4 Identification and analytical credibility (what makes the results persuasive)

4.1 What exactly drives the results

The paper's strongest results require three ingredients together:

- **portfolio-level attention allocation:** investors process more information about factors that matter for many assets at once;
- **overconfidence:** investors overestimate the precision of the information they process;
- **dynamic asset pricing:** information affects the timing of uncertainty resolution and therefore affects return volatility and predictability.

Interpretation. Limited attention alone gives category learning. Overconfidence converts category learning into excess comovement and return predictability.

4.2 Why limited attention alone is not enough

A fully rational investor with $\phi = 1$ does not produce the main excess-volatility result. More information either leaves return variance unchanged when discounting is zero or lowers it when discounting is positive.

Interpretation. The paper is explicit that information frictions by themselves are not enough. Behavioral overreaction is necessary for the main asset-pricing predictions.

4.3 Why the representative-investor setup still matters

The model uses a representative investor rather than many heterogeneous investors. The authors argue that retail investors often face similar information sources, similar attention constraints, and similar biases. Therefore their errors need not average out in prices.

Interpretation. The representative-investor shortcut is not meant to say investors are literally identical; it is meant to capture correlated learning errors generated by common media, common reports, and common psychological frictions.

4.4 Why the no-learning-from-prices assumption is important

The paper does not let the investor infer fundamentals from prices. The authors argue that learning from prices is not free: it requires knowing the information sets, motives, and trades of other market participants. For retail investors, that is especially difficult.

Interpretation. This assumption shuts down a force that would partially offset attention constraints. It makes the paper's learning channel sharper and easier to analyze.

4.5 What is empirically testable

The theory yields several predictions that can be mapped to data:

- return correlation can exceed fundamental correlation;
- sectors with lower average return correlation should have more informative prices;
- improvements in information technology should lower average stock-return correlation over time;
- after controlling for overconfidence, sectors with lower return correlation should show stronger overreaction-based predictability;

- ignored public signals should predict returns more in sectors with higher return correlation.

Interpretation. The model is theoretical, but it speaks directly to measured synchronicity, price informativeness, reversals, momentum, and accounting-based return predictors.

4.6 What the paper does not fully pin down

- It does not microfound media production, analyst coverage, or investor heterogeneity.
- It does not explain exactly why arbitrage fails beyond relying on standard limits-to-arbitrage logic.
- It does not estimate the size of the attention constraint or the overconfidence parameter in data.
- It abstracts from persistence in fundamentals, which helps analytically but removes some richer dynamics.

Interpretation. The paper is best viewed as a mechanism paper: it shows one coherent way that limited attention and overconfidence can produce a set of puzzling facts together.

5 Tables and figures

5.1 Figure D.1: return-variance amplification as a function of processed information

Read:

- The curve is hump-shaped rather than monotone.
- For low to moderate information levels, more processed information raises return-variance amplification.
- The vertical dashed line marks the threshold $\bar{x}$. To the left of it, overreaction dominates; to the right of it, the pure precision effect dominates.
- The paper restricts its main analysis to the empirically plausible region in which individual investors operate below this threshold.

Economic message. This figure is the numerical backbone behind Proposition 2B. It shows why more information need not stabilize prices when investors are overconfident about the quality of their own processing.

5.2 Table D.1: thresholds for the variance-amplification region

Read:

- For $r = 5\%$ and $\phi = 2$, the threshold is about $\bar{x} = 0.521$.
- That corresponds to roughly a 40.6% reduction in the variance of investor belief.

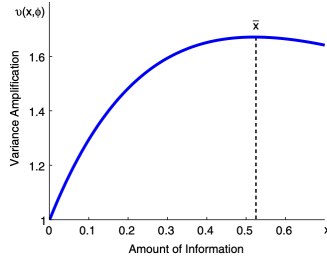


Fig. D.1. The relation between return variance amplification, $v(x, \phi)$, and the amount of processed information, x . In the figure, the investor's overconfidence parameter ϕ is 2 and the interest rate r is 5%.

Table D.1
The threshold in amount of information

ϕ	$\bar{x}$	$\Delta\sigma^2/\sigma^2$ (%)
<i>Panel A. $r = 0$</i>		
1.2	0.857	-57.6
1.4	0.751	-52.8
1.6	0.669	-48.8
1.8	0.603	-45.3
2.0	0.549	-42.2
2.2	0.504	-39.6
2.4	0.466	-37.2
<i>Panel B. $r = 5\%$</i>		
1.2	0.646	-47.6
1.4	0.657	-48.2
1.6	0.613	-45.8
1.8	0.565	-43.2
2.0	0.521	-40.6
2.2	0.483	-38.3
2.4	0.449	-36.2

The table provides the values of the threshold ($\bar{x}$) in the amount of information and the corresponding percentage reduction in the variance of investor belief ($\Delta\sigma^2/\sigma^2$) for different values of the investor's overconfidence parameter (ϕ) and the discount rate (r).

- Across the table's parameter values, the threshold remains fairly high for an individual investor's own learning process.
- The required overconfidence threshold in Proposition 2 is modest; the paper stresses that the main restriction is really about the amount of self-processed information being below $\bar{x}$.

Economic message. The table argues that the empirically relevant region for individual investors is exactly the one where extra information can increase return variance through overreaction.

5.3 Summary table: what each proposition adds

Result	Key condition	Economic implication
Proposition 1	Common factors matter enough for the portfolio; attention is scarce	Investors learn more about market and sector categories than about individual firms.
Proposition 2	Overconfidence is high enough and processed information is below the threshold	More processed information can raise return variance because prices overreact to noisy signals.
Proposition 3	Category learning plus Proposition 2B	Return correlation exceeds fundamental correlation: <i>excess comovement</i> .
Proposition 4	Some sectors have higher information efficiency (y_i)	Lower synchronicity and higher price informativeness go together.
Proposition 5	Information technology improves over time	Relative firm-specific volatility rises and average return correlation falls.
Propositions 6–7	Attention to firms differs across sectors	Lower-correlation sectors have stronger overreaction-based predictability, while ignored signals predict less.

Table 1: Compact map of the paper's core propositions and their economic meaning.

Economic message. The propositions are tightly linked. Proposition 1 gives the information structure; Proposition 2 gives the overreaction technology; the rest translate those two ingredients into comovement, informativeness, and predictability implications.

6 Short summary

- The paper models attention as a scarce cognitive resource and lets investors choose how to allocate it across market, sector, and firm-specific factors.
- Optimal attention allocation creates category learning: investors process relatively more common information than firm-specific information.
- Overconfidence makes investors overestimate the precision of the information they process, which makes them overreact to category-level signals.
- This combination generates excess return comovement, a negative relation between synchronicity and price informativeness, a decline in correlation as information technology improves, and new cross-sectional predictions for return predictability.
- The paper's core economic claim is that the market's information structure is endogenous to investors' attention choices, and that this endogenous structure helps explain several puzzles in stock-return data at once.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that limited attention does not merely slow information diffusion. It changes the *composition* of the information that gets processed. Investors optimally tilt toward common factors, thereby learning about securities in categories. When this endogenous category learning is combined with overconfidence, the model produces excess comovement, lower informativeness in high-correlation sectors, declining correlation as information technology improves, and sharp cross-sectional implications for return predictability.

7.2 Economic intuition

If one unit of attention can help with many stocks at once, it is optimal to spend that unit on market or sector information rather than on one firm's details. This makes common information more prominent in prices. Overconfidence then magnifies the pricing impact of exactly that common information. The result is too much movement together and too little firm-level distinction in sectors where attention remains coarse.

7.3 Takeaway

The paper's lasting contribution is to connect **attention allocation**, **behavioral bias**, and **asset-pricing moments**. Comovement is not just a reflection of common fundamentals; it is also a reflection of the way investors economize on cognition.